Dylan Becchetti

Mechanical Engineer | Manufacturing Automation & Engineering Software

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Portfolio: formsby-systems.github.io/engineering-portfolio

PROFILE

Lehigh mechanical engineering graduate with three Cardinal Glass manufacturing internships and independent work in research software and CAD. Available January 2027 for full-time U.S. remote work.

EXPERIENCE

Cardinal Glass Industries | St. Louis Park, MN

Continuous Improvement Engineering Intern, summers 2024–2025 | Engineering Intern, summer 2023

- Worked with engineers and operators to define shift handoff needs, then built an Excel/VBA tracker that combined R&D production files into a daily part-status view used in debriefs.
- Wrote illustrated assembly instructions, planned spare-parts storage and rack signage across 14 locations, and measured cleanroom airflow with a hood and micromanometer.
- Inspected pilot-line equipment, analyzed data in Minitab, drafted AutoCAD layouts and contributed to design failure mode and effects analysis.

SELECTED PROJECTS

Wilbur Powerhouse | Mechanical Design, year-long team capstone

- Built the full-scale OpenSCAD assembly. Widened the ring gear to 79.5 inches and moved the drive while preserving the inner tooth profile.
- Worked with the team on carrier FEA and CNC fabrication. Identified a bore clearance change during fit checks on the first welded gear unit.

Midas Futures | Creator & Technical Lead

- Led a year of development on a Python research and Practice platform with modular strategies, reproducible batch replay, checkpointed ablations and Monte Carlo tests.
- Unified historical and forward rules, matching 228 session decisions and 225 trade lifecycles against a sealed reference. Reproduced 11 decision files byte for byte in a separate Railway run.
- Built a research dashboard that shows account fees and modeled cash separately from trading P&L. The installed dashboard displayed 1,410 modeled account cases.

Operalith | Founder & Product Lead, independent project

- Built a configurable Next.js workspace where users choose tools and features while keeping records connected. Designed account saves that prevent duplicate writes and reject stale edits. Verified that undo preserved 11 prior history entries.

Formsby | Creator, search prototype

- Built a Next.js/PostgreSQL housing search prototype with queued jobs and private ranked results. Kept results tied to the latest run and made queue failures visible.

EDUCATION & TOOLS

Lehigh University | B.S. Mechanical Engineering, May 2026

Materials Science & Engineering minor expected Dec. 2026 | SOLIDWORKS Associate (CSWA), 2024.

Python, JavaScript, React/Next.js, SQL/PostgreSQL, Git, automated testing, OpenSCAD, SOLIDWORKS, AutoCAD, Excel/VBA, Minitab.